Health-related quality of life and pain outcomes with [^{177}Lu]Lu-PSMA-617 plus standard of care versus standard of care in patients with metastatic castration-resistant prostate cancer (VISION): a multicentre, open-label, randomised, phase 3 trial

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Summary

Background In VISION, the prostate-specific membrane antigen (PSMA)-targeted radioligand therapy lutetium-177 [^{177}Lu]Lu-PSMA-617 (vipivotide tetraxetan) improved radiographic progression-free survival and overall survival when added to protocol-permitted standard of care in patients with metastatic castration-resistant prostate cancer. Here, we report additional health-related quality of life (HRQOL), pain, and symptomatic skeletal event results.

Methods This multicentre, open-label, randomised, phase 3 trial was conducted at 84 cancer centres in nine countries in North America and Europe. Eligible patients were aged 18 years or older; had progressive PSMA-positive metastatic castration-resistant prostate cancer; an Eastern Cooperative Oncology Group (ECOG) performance status score of 0–2; and had previously received of at least one androgen receptor pathway inhibitor and one or two taxane-containing regimens. Patients were randomly assigned (2:1) to receive either [^{177}Lu]Lu-PSMA-617 plus protocol-permitted standard of care ([^{177}Lu]Lu-PSMA-617 group) or standard of care alone (control group) using permuted blocks. Randomisation was stratified by baseline lactate dehydrogenase concentration, liver metastases, ECOG performance status, and androgen receptor pathway inhibitor inclusion in standard of care. Patients in the [^{177}Lu]Lu-PSMA-617 group received intravenous infusions of 7·4 gigabecquerel (GBq; 200 millicurie [mCi]) [^{177}Lu]Lu-PSMA-617 every 6 weeks for four cycles plus two optional additional cycles. Standard of care included approved hormonal treatments, bisphosphonates, and radiotherapy. The alternate primary endpoints were radiographic progression-free survival and overall survival, which have been reported. Here we report the key secondary endpoint of time to first symptomatic skeletal event, and other secondary endpoints of HRQOL assessed with the Functional Assessment of Cancer Therapy-Prostate (FACT-P) and EQ-5D-5L, and pain assessed with the Brief Pain Inventory-Short Form (BPI-SF). Patient-reported outcomes and symptomatic skeletal events were analysed in all patients who were randomly assigned after implementation of measures designed to reduce the dropout rate in the control group (on or after March 5, 2019), and safety was analysed according to treatment received in all patients who received at least one dose of treatment. This trial is registered with ClinicalTrials.gov, NCT03511664, and is active but not recruiting.

Findings Between June 4, 2018, and Oct 23, 2019, 831 patients were enrolled, of whom 581 were randomly assigned to the [^{177}Lu]Lu-PSMA-617 group (n=385) or control group (n=196) on or after March 5, 2019, and were included in analyses of HRQOL, pain, and time to first symptomatic skeletal event. The median age of patients was 71 years (IQR 65–75) in the [^{177}Lu]Lu-PSMA-617 group and 72·0 years (66–76) in the control group. Median time to first symptomatic skeletal event or death was 11·5 months (95% CI 10·3–13·2) in the [^{177}Lu]Lu-PSMA-617 group and 6·8 months (5·2–8·5) in the control group (hazard ratio [HR] 0·50, 95% CI 0·40–0·62). Time to worsening was delayed in the [^{177}Lu]Lu-PSMA-617 group versus the control group for FACT-P score (HR 0·54, 0·45–0·66) and subdomains, BPI-SF pain intensity score (0·52, 0·42–0·63), and EQ-5D-5L utility score (0·65, 0·54–0·78). Grade 3 or 4 haematological adverse events included decreased haemoglobin (80 [15%] of 529 assessable patients who received [^{177}Lu]Lu-PSMA-617 plus standard of care *vs* 13 [6%] of 205 who received standard of care only), lymphocyte concentrations (269 [51%] *vs* 39 [19%]), and platelet counts (49 [9%] *vs* five [2%]). Treatment-related adverse events leading to death occurred in five (1%) patients who received [^{177}Lu]Lu-PSMA-617 plus standard of care (pancytopenia [n=2], bone marrow failure [n=1], subdural haematoma [n=1], and intracranial haemorrhage [n=1]) and no patients who received standard of care only.

Interpretation [^{177}Lu]Lu-PSMA-617 plus standard of care delayed time to worsening in HRQOL and time to skeletal events compared with standard of care alone. These findings support the use of [^{177}Lu]Lu-PSMA-617 in patients with metastatic castration-resistant prostate cancer who received previous androgen receptor pathway inhibitor and taxane treatment.

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Introduction

Despite substantial therapeutic advances since 2010, metastatic castration-resistant prostate cancer remains an incurable and fatal disease.[1,2] In the past two decades, several new classes of therapy shown to provide a survival benefit in patients with metastatic castration-resistant prostate cancer have been approved. These include taxanes (docetaxel and cabazitaxel), androgen receptor pathway inhibitors (such as abiraterone acetate and enzalutamide), immunotherapy (sipuleucel-T), and bone-targeted radionuclide therapy (radium-223).[1]

Patients with progressive metastatic castration-resistant prostate cancer who previously received androgen receptor pathway inhibitors and taxanes have few treatment options and are at high risk of impaired health-related quality of life (HRQOL), morbidity, and mortality.[3] Treatment of this type of cancer is associated with pronounced physical, emotional, and functional wellbeing impairment;[4] therefore, HRQOL is an important outcome to consider in treatment decision making for patients with metastatic castration-resistant prostate cancer, in addition to efficacy, safety, and tolerability considerations.

Symptomatic skeletal events (such as fracture, spinal cord compression, and pain) have a pronounced negative effect on HRQOL in patients with bone metastasis.[5,6] Up to 90% of patients with metastatic castration-resistant prostate cancer have bone metastases,[7] and bone-targeted therapies such as bisphosphonates (zoledronic acid) or denosumab are key for the prevention of symptomatic skeletal events.[1,2] Therefore, delaying the time to symptomatic skeletal events, regardless of bone-targeted therapy use, is an important aspect of anticancer therapy efficacy in patients with metastatic castration-resistant prostate cancer and bone metastasis.

Prostate-specific membrane antigen (PSMA) is a transmembrane glutamate carboxypeptidase with highly upregulated expression in prostate cancer cells.[8] Radioligand therapy selectively targets cancer cells via specific cell-surface proteins and spares most healthy tissues.[9] Lutetium-177 [^{177}Lu]Lu-PSMA-617 (vipivotide tetraxetan) is a high affinity PSMA-targeted small

Research in context

Evidence before this study

Lutetium-177 [^{177}Lu]Lu-PSMA-617 is a small molecule targeted radioligand therapy for the treatment of patients with metastatic castration-resistant prostate cancer. We searched PubMed for publications in English from Jan 1, 2016, to Sept 30, 2021, using the search terms "metastatic castration-resistant prostate cancer" AND "prostate-specific membrane antigen" AND "lutetium" AND "clinical trial". We found three small, phase 1–2 trials of [^{177}Lu]Lu-PSMA-617 in patients with metastatic castration-resistant prostate cancer. Of these, only one was a randomised controlled trial—the phase 2 TheraP study—in which [^{177}Lu]Lu-PSMA-617 improved prostate-specific antigen response compared with cabazitaxel for patients with metastatic castration-resistant prostate cancer who previously received androgen receptor pathway inhibitors or chemotherapy, or both.

Added value of this study

To our knowledge, VISION is the first phase 3 study of [^{177}Lu]Lu-PSMA-617, in which we previously showed longer radiographic progression-free survival and overall survival when [^{177}Lu]Lu-PSMA-617 was added to protocol-permitted standard of care in patients with metastatic castration-resistant prostate cancer who were positive for prostate-specific membrane antigen (PSMA) and who had previously received at least one androgen receptor pathway inhibitor and one or two taxanes. The optimal treatment aims not only to extend life and delay disease progression but also to improve or maintain health-related quality of life (HRQOL). Patients with metastatic castration-resistant prostate cancer receiving guideline-directed therapy with androgen receptor pathway inhibitors or taxanes have a high risk of impaired HRQOL, morbidity, and mortality. Bone metastases are common in these patients, and symptomatic skeletal events have a substantial negative impact. VISION data show that [^{177}Lu]Lu-PSMA-617 plus standard of care delayed the time to worsening of patient-reported HRQOL and pain in multiple domains compared with standard of care alone. In post-hoc analyses, this finding was extended from the prespecified composite outcome of worsening in Functional Assessment of Cancer Therapy-Prostate, EQ-5D-5L, and Brief Pain Inventory-Short Form scores or clinical disease progression or death to simply worsening in scores, with similar results. Furthermore, the time to first symptomatic skeletal event was also delayed in patients who received [^{177}Lu]Lu-PSMA-617 plus standard of care versus standard of care alone, regardless of use of bone-targeted therapy.

Implications of all the available evidence

These new findings add to the evidence that addition of [^{177}Lu]Lu-PSMA-617 to standard of care results in longer radiographic progression-free survival and overall survival, with no adverse effects on HRQOL or pain, in patients with metastatic castration-resistant prostate cancer. Patient-reported outcomes also improved or were similar with [^{177}Lu]Lu-PSMA-617 versus cabazitaxel in the TheraP trial. Findings from the phase 3 VISION studies and previous phase 2 studies provide a strong rationale for use of [^{177}Lu]Lu-PSMA-617 as a treatment option in patients with advanced metastatic castration-resistant prostate cancer.

molecule radioligand therapy that delivers β particle radiation specifically to PSMA-expressing cells and the surrounding microenvironment.[10]

The kidneys and bone marrow are recognised as dose-limiting organs for therapeutic radiopharmaceuticals. In patients receiving [^{177}Lu]Lu-PSMA-617, the kidneys are exposed to radiation because urinary excretion is the principal route of drug elimination and PSMA is expressed in proximal tubular cells. The bone marrow is at risk of toxicity because of radiation-induced myelosuppression. Patients with severe haematological and renal complications could require interventions that negatively affect quality of life.

In the phase 2 TheraP trial,[11] the prostate-specific antigen (PSA) response rate was higher in patients who received [^{177}Lu]Lu-PSMA-617 than in those who received cabazitaxel, and a smaller proportion of patients had grade 3 or 4 adverse events in the [^{177}Lu]Lu-PSMA-617 group. For the alternate primary endpoints in the phase 3 VISION trial,[12] [^{177}Lu]Lu-PSMA-617 plus standard of care showed improved radiographic progression-free survival (hazard ratio [HR] 0·40, 99·2% CI 0·29–0·57; p<0·001) and overall survival (0·62, 0·52–0·74) compared with protocol-permitted standard of care alone in patients with advanced PSMA-positive metastatic castration-resistant prostate cancer.

Top-level prespecified HRQOL and pain results from VISION have been briefly reported previously, with time to worsening in the Functional Assessment of Cancer Therapy-Prostate (FACT-P)[13] total score (HR 0·54, 95% CI 0·45–0·66) and Brief Pain Inventory-Short Form (BPI-SF)[14] pain intensity score (0·52, 0·42–0·63) favouring the addition of [^{177}Lu]Lu-PSMA-617 to standard of care.[12] These analyses included clinical disease progression or death as a composite endpoint together with HRQOL and pain deterioration.

Here, we report prespecified and post-hoc non-composite analyses of HRQOL and pain and new results on symptomatic skeletal events, haematological variables, and renal variables, as well as further assessing the safety of [^{177}Lu]Lu-PSMA-617 in patients with progressive PSMA-positive metastatic castration-resistant prostate cancer who previously received at least one androgen receptor pathway inhibitor and one or two taxane-containing regimens, as described previously.[12]

Methods

Study design and participants

VISION was a multicentre, open-label, randomised, phase 3 trial conducted at 84 centres in nine countries in North America and Europe (81 centres had at least one patient randomly assigned to the [^{177}Lu]Lu-PSMA-617 group; appendix pp 20–21). Eligible patients were aged 18 years or older; had progressive metastatic castration-resistant prostate cancer (defined as at least one PSMA-positive metastatic lesion on baseline CT, MRI, or bone scan imaging and no PSMA-negative lesions, as per protocol criteria);[12] an Eastern Cooperative Oncology Group (ECOG) performance status score of 0–2; life expectancy of at least 6 months; adequate hepatic, renal, and bone marrow function; and had previously received at least one androgen receptor pathway inhibitor and one or two taxane-containing regimens. PSMA-positive tumour status was measured using gallium-68 [^{68}Ga]Ga-PSMA-11 (gozetotide) PET-CT imaging during screening and evaluated by independent central review. Radiographic imaging was evaluated in accordance with Response Evaluation Criteria in Solid Tumours (version 1.1) and Prostate Cancer Working Group 3 criteria. Patients who were candidates for additional chemotherapy at screening or required treatments that were not permitted under the study protocol were ineligible.[12] The protocol is available online. All participants provided written informed consent. This study was conducted in accordance with the Declaration of Helsinki, Good Clinical Practice guidelines, and any applicable local regulations.

For the **protocol** see https://clinicaltrials.gov/ProvidedDocs/64/NCT03511664/Prot_000.pdf

Randomisation and masking

Patients were randomly assigned (2:1) with an interactive response system (Endpoint Clinical, San Francisco, CA, USA) to receive either [^{177}Lu]Lu-PSMA-617 plus protocol-permitted standard of care ([^{177}Lu]Lu-PSMA-617 group) or standard of care alone (control group) using permuted blocks (block size not disclosed to ensure concealment). The investigators enrolled patients at each site. Randomisation was stratified by baseline lactate dehydrogenase concentration (≤260 IU/L *vs* >260 IU/L), presence of liver metastases (yes *vs* no), ECOG performance status (0–1 *vs* 2), and androgen receptor pathway inhibitor included in protocol-permitted standard of care (yes *vs* no). The randomisation list was generated and approved by the initial study statistician who was not further involved in the trial. Neither patients nor investigators were masked to treatment assignment due to the nature of the intervention.

Procedures

In addition to standard of care, patients in the [^{177}Lu]Lu-PSMA-617 group received intravenous infusions of 7·4 gigabecquerel (GBq; 200 millicurie [mCi]) [^{177}Lu]Lu-PSMA-617 every 6 weeks for four cycles plus two optional additional cycles (up to six in total) that could be administered in patients who tolerated therapy and showed evidence of response (eg, radiological, PSA, clinical benefit), at investigator's discretion. One dose of [^{177}Lu]Lu-PSMA-617 could also be reduced or delayed by up to 4 weeks.[12]

Tumour assessments (CT, MRI, and bone scan imaging) were done every 8 weeks for the first 24 weeks, then every 12 weeks until end of treatment. Standard of care with or without [^{177}Lu]Lu-PSMA-617 continued until radiographic disease progression, unacceptable toxicity, absence of clinical benefit, non-compliance

See **Online** for appendix

with protocol, or until a prohibited treatment was deemed necessary. Patients could also be withdrawn from the study after withdrawal of consent or at the discretion of the sponsor or investigator. Patients who completed [^{177}Lu]Lu-PSMA-617 continued to receive standard of care. Investigator-determined standard of care included approved hormonal treatments (such as abiraterone and enzalutamide), bisphosphonates, radiotherapy, denosumab, glucocorticoids, hydration, and analgesics. Cytotoxic chemotherapy, systemic radioisotopes (eg, radium-223), immunotherapy, and investigational drugs were not allowed.[12]

Patient-reported outcomes (PROs) included the FACT-P, BPI-SF, and EQ-5D-5L instruments.[15] Questionnaires were completed electronically by patients or with assistance during a face-to-face interview at baseline (before or at randomisation), on the first day of each cycle, and at the end of treatment. The cycle length was 6 weeks for cycles one to six (standard of care with or without [^{177}Lu]Lu-PSMA-617) and 12 weeks for cycle seven onwards (continuation of standard of care). The FACT-P consists of the Functional Assessment of Cancer Therapy-General (27 items) and the Prostate Cancer Subscale (12 items). The FACT-P score (range 0–156) is the sum of scores (39 items), with higher scores indicating better HRQOL. FACT-P subscales are physical wellbeing, social or family wellbeing, emotional well-being, functional wellbeing, prostate cancer subscale, and pain-related subscale. The trial outcome index is a composite of physical wellbeing, functional wellbeing, and prostate cancer subscales. The EQ-5D-5L generates a preference-based health-state utility score (EQ-5D utility index; range –0·594 to 1) consisting of five dimensions (mobility, self-care, usual activities, pain or discomfort, and anxiety or depression) and an overall health-state score based on a visual analogue scale (EQ-5D VAS; range 0–100). The BPI-SF assesses pain intensity (range 0–10; no pain to worse pain) and how pain interferes with daily activities (range 0–10; no interference to complete interference). BPI-SF pain intensity is a mean of four individual scales (worst pain intensity, least pain intensity, mean pain intensity, and current pain); we planned to analyse time to worsening in worst pain intensity only.

Haematology and clinical chemistry variables were assessed every week during cycle 1, every 2 weeks during cycles 2–6, every 12 weeks after cycle 6, and at end of treatment. Haematological adverse events were monitored during treatment and were defined as occurring from first administration up to 30 days after the last dose of study treatment or before subsequent anticancer treatment, whichever occurred first. Symptomatic skeletal events were monitored during the study and until the end of treatment.

Adverse events and haematology were recorded based on Common Terminology Criteria for Adverse Events (CTCAE) grading (version 5.0).

Outcomes

The alternate coprimary endpoints were radiographic progression-free survival (defined as time from the date of randomisation to radiographic disease progression) and overall survival (defined as the time from randomisation to death from any cause), and were reported previously.[12] Our original protocol only stated overall survival as the primary endpoint, but a protocol amendment was made on July 8, 2019, following consultation with the US Food and Drug Administration to add imaging-based progression-free survival as an alternate primary endpoint, meaning that the trial would be deemed to be positive if either or both of the primary endpoints were significant.

Key secondary endpoints were time to first symptomatic skeletal event or death (defined as the time from randomisation to first new pathological bone fracture, spinal cord compression, tumour-related orthopaedic surgical intervention, requirement for radiotherapy to relieve bone pain, or death from any cause, whichever occurred first), overall response rate, and disease control rate (both measured by RECIST version 1.1 criteria). Safety of [^{177}Lu]Lu-PSMA-617, patient-reported HRQOL and pain outcomes (assessed by FACT-P, EQ-5D-5L, and BPI-SF), health economics, progression-free survival (defined as time from date of randomisation to first evidence of radiographic progression, clinical progression, PSA progression, or death from any cause, whichever occurred first), and biochemical response (including PSA response, defined as ≥50% decrease from baseline, and alkaline phosphatase and lactate dehydrogenase concentrations) were additional secondary endpoints. Top-line analyses of objective response rate, disease control rate, safety, HRQOL and pain outcomes, and PSA response were reported previously.[12]

Definitions of change from baseline in FACT-P and BPI-SF were based on established minimum clinically important differences of 6–10 points for FACT-P total score, 1–2 points for FACT-P pain-related subscale, 5–9 points for FACT-P trial outcome index, 2–3 points for FACT-P other subscales, and an increase of at least 30% or at least 2 points for BPI-SF scales and subscales.[16,17] Time to worsening in FACT-P was predefined in our statistical analysis plan as the time from randomisation to a decrease of at least 10 points from baseline for total score, at least 2 points from baseline for pain-related subscale, at least 9 points from baseline for trial outcome index, or at least 3 points from baseline for other subscales. Time to worsening in EQ-5D-5L utility score was predefined as the time from randomisation to any decrease or no change from baseline. Time to worsening in BPI-SF scales and subscales was predefined as the time from randomisation to an increase of at least 30% or at least 2 points from baseline.

Statistical analysis

We estimated that a sample size of 814 patients would provide 84% power to detect a HR of 0·67

for radiographic progression-free survival (one-sided α=0·004) and 90% power to detect a HR for overall survival of 0·73 (one-sided α=0·02), as previously described.[12] The null hypotheses for the alternate primary endpoints were that there was no difference in overall survival and radiographic progression-free survival between the [177Lu]Lu-PSMA-617 and control groups. The null hypothesis for the key secondary endpoint was that there was no difference in time to first symptomatic skeletal event between treatment groups. There was no hypothesis testing for PROs and no α control was applied.

In the analyses presented in this Article, PROs and symptomatic skeletal events were analysed according to assigned treatment group in the progression-free survival full analysis set (PFS-FAS), defined as patients who were randomly assigned after implementation of measures designed to reduce the dropout rate in the control group (on or after March 5, 2019). The decision to analyse these endpoints in this population was made after we observed a high incidence of withdrawal in the control group, which could have affected interpretability of the results. As a result, enhanced trial-site education measures were implemented on March 5, 2019, following discussion with regulatory authorities. The withdrawal rate after these measures were implemented was substantially lower than before March 5, 2019; therefore, the analysis of these key secondary endpoints was amended to include only patients randomly assigned to treatment from March 5, 2019, onwards. Details of these measures have been published elsewhere.[12] Safety was analysed according to the treatment received in the safety analysis set, defined as all patients who received at least one dose, regardless of random group assignment. Overall survival was analysed in the full analysis set, defined as all patients who underwent randomisation (before and after March 5, 2019), and all other endpoints were analysed in the PFS-FAS.

In prespecified analyses of BPI-SF, FACT-P, and EQ-5D-5L, the endpoint was the composite of worsening in score, clinical disease progression (excluding radiographic and PSA progression), or death due to any cause.[12] Clinical disease progression was assessed by investigators on the basis of marked escalation in cancer-related pain, immediate need for initiation of new anticancer treatment, surgical or radiological intervention, or marked deterioration in ECOG performance status higher or equal to grade 3. In post-hoc analyses of PROs, only worsening in score was considered an event and time to worsening in EQ-5D-5L was defined as the time from randomisation to a decrease in utility score of at least 0·10 points from baseline,[18] and time to worsening in BPI-SF scales was defined as the time from randomisation to an increase of at least 2 points from baseline. Prespecified analyses for time to improvement after worsening in BPI-SF pain intensity scale was defined as the time from worsening to occurrence of a score lower than or equal to the baseline value. Linear mixed effects models were used to analyse treatment differences in changes from baseline in PROs.

For prespecified analyses of time to first symptomatic skeletal event and post-hoc subgroup analyses of time to first symptomatic skeletal event (with or without inclusion of death) by whether patients received bone-targeted agents, HRs and 95% CIs were estimated using the stratified Cox regression model.[12] Median, percentiles, and 95% CIs were estimated using the Kaplan-Meier method. For PROs, the prespecified method of statistical

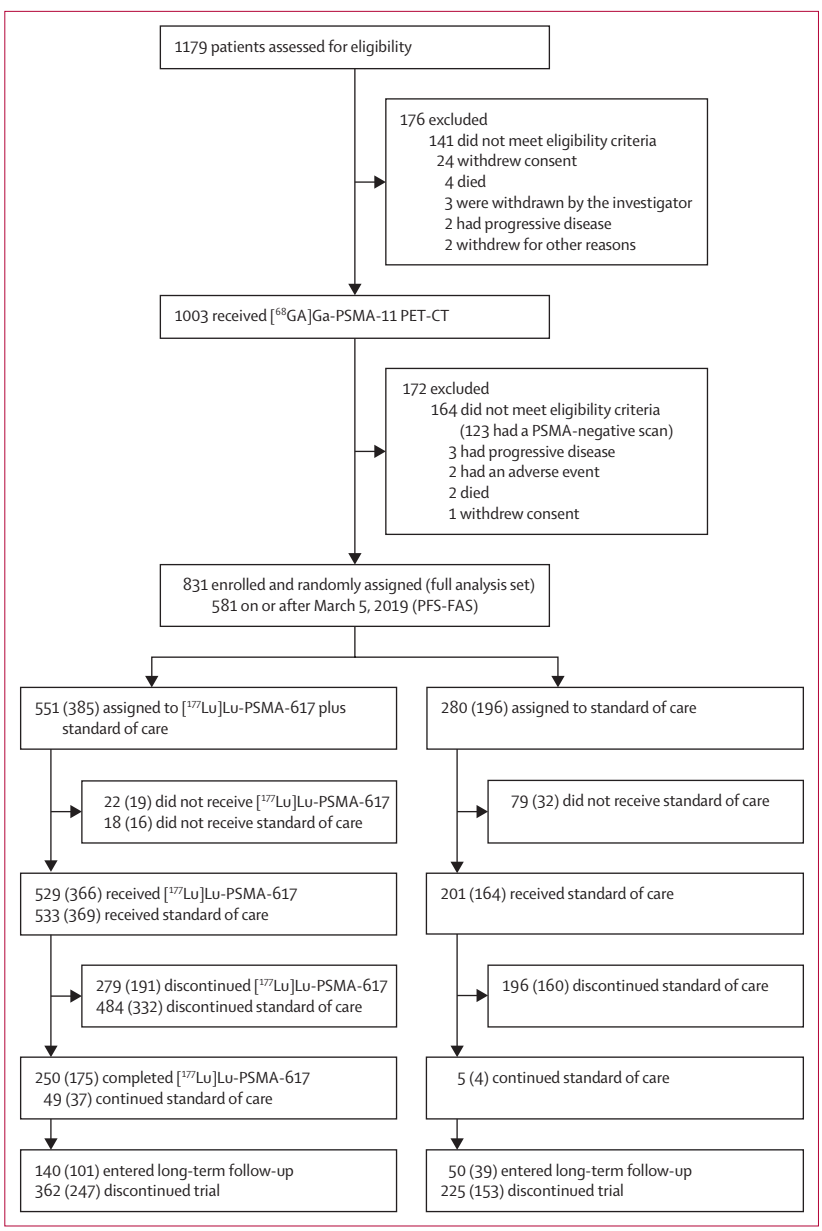
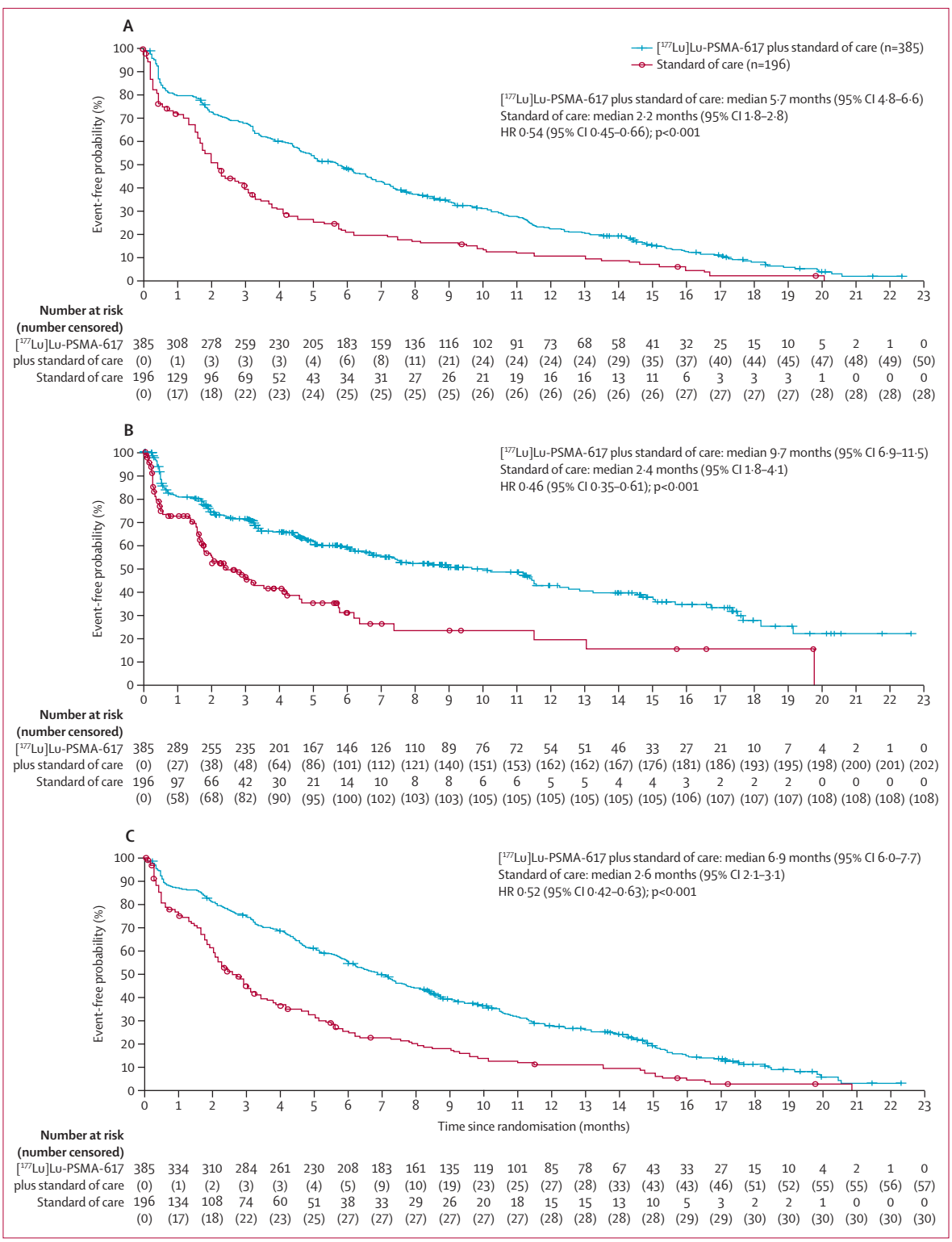


Figure 1: **Trial profile**
Numbers in parentheses represent the number of participants who underwent randomisation on or after March 5, 2019 (PFS-FAS). Ga=gallium. Lu=lutetium. PFS-FAS=progression-free survival full analysis set. PSMA=prostate-specific membrane antigen.

comparison was the Wald χ^2 test from the stratified Cox proportional hazards model, stratified by baseline lactate dehydrogenase concentration (≤260 U/mL *vs* >260 U/mL), presence of liver metastases (yes *vs* no), ECOG performance status (0–1 *vs* 2), and inclusion of androgen receptor pathway inhibitor in protocol-permitted standard of care at



(Figure 2 continues on next page)

randomisation (yes *vs* no). The Cox model was also used to estimate HRs and 95% CIs. Median, percentiles, and 95% CIs were estimated using the Kaplan-Meier method. The Cox proportional hazards assumption was tested by visual inspection for the primary endpoints. Secondary endpoints were assessed at the nominal 5% α level.

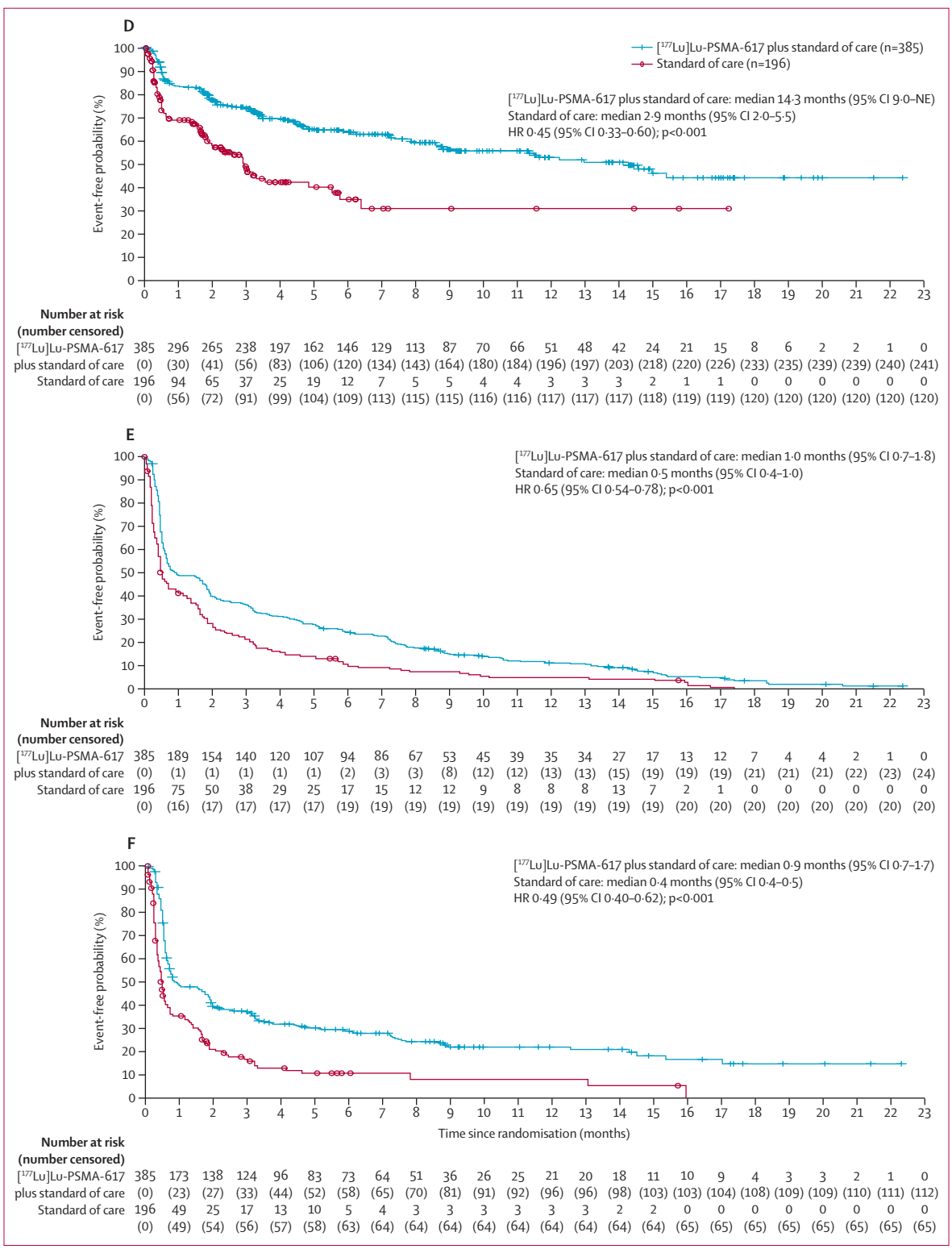


Figure 2: **Time to worsening in FACT-P score, BPI-SF scale, and EQ-5D-5L utility score**
Analyses in 581 patients randomly assigned on or after March 5, 2019 (PFS-FAS). Time to worsening of 10 points or more in the FACT-P score, clinical disease progression, or death (prespecified analysis; A) or in the FACT-P score (post-hoc analysis; B). Time to worsening of at least 30% or at least 2 points in the BPI-SF scale, clinical disease progression, or death (prespecified; C) or an increase of at least 2 points in the BPI-SF scale (post hoc; D). Time to worsening in EQ-5D-5L utility score, defined as the time to any decrease in utility score relative to baseline; or no change in score from baseline (prespecified; E) and a decrease in score of at least 0·10 points from baseline (post hoc; F). BPI-SF=Brief Pain Inventory-Short Form. FACT-P=Functional Assessment of Cancer Therapy-Prostate. HR=hazard ratio. Lu=lutetium. NE=not estimable. PSMA=prostate-specific membrane antigen.

HRQOL and pain time to worsening were excluded from study-wide type I error control and a prespecified analysis was done using an unadjusted two-sided α of 0·05; no sensitivity analyses were prespecified or performed. Multiplicity was uncontrolled in post-hoc analyses. All p values are nominal, descriptive, and non-inferential. Adverse events and laboratory measurements were summarised by descriptive statistics, and prespecified assessments of change in haemoglobin levels were conducted.

Patients without an event were censored at the date of the last PRO assessment; patients without evaluable data were censored at randomisation. The approach to missing items for each HRQOL and pain scale is detailed in the statistical analysis plan, according to the instrument developer's guidelines. All items for EQ-5D-5L, more than 80% of items for FACT-P, and at least three of four items for BPI-SF need to have been completed for a participant to be considered evaluable. No methods for imputation of missing data were prespecified. All statistical analyses were done using SAS (version 9.4). This trial is registered with ClinicalTrials.gov, NCT03511664.

For the **statistical analysis plan** see https://clinicaltrials.gov/ProvidedDocs/64/NCT03511664/SAP_001.pdf

Role of the funding source

The funder of the study had a role in study design, data analysis, data interpretation, and writing of the report, but had no role in data collection.

Results

Between June 4, 2018, and Oct 23, 2019, 831 patients were enrolled, of whom 581 were randomly assigned to the [^{177}Lu]Lu-PSMA-617 group (n=385) or control group (n=196) on or after March 5, 2019 (the PFS-FAS; figure 1).[12] Safety analyses were done in 734 patients who received at least one dose of study treatment (529 in the [^{177}Lu]Lu-PSMA-617 group and 205 in the control group [four patients randomly assigned to the [^{177}Lu]Lu-PSMA-617 group only received standard of care so were included in the control group for the safety analyses]).[12] Baseline characteristics were balanced between the groups (appendix p 2). The median duration of exposure to treatment has been previously reported.[12] The median age of patients in the PFS-FAS was 71 years (IQR 65–75) in the [^{177}Lu]Lu-PSMA-617 group and 72 years (66–76) in the control group. The majority of patients were White (336 [87%] of 385 in the [^{177}Lu]Lu-PSMA-617 group and 166 [85%] of 196 in the control group; appendix p 2).

Among patients remaining on study treatment, questionnaire completion rates were high (appendix p 3). Among patients remaining in the study, PRO completion rates were similar in the [^{177}Lu]Lu-PSMA-617 group and the control group during study treatment (appendix p 3). However, the study dropout rate was higher in the control group than in the [^{177}Lu]Lu-PSMA-617 group (247 [64%] of 385 in the [^{177}Lu]Lu-PSMA-617 group *vs* 153 [78%] of 196 in the control group). Questionnaire completion rates were similar at the end-of-treatment visit, with data available for 175 (49%) of 354 patients in the [^{177}Lu]Lu-PSMA-617 group remaining in the study and 87 (45%) of 192 patients in the control group remaining in the study for FACT-P, 177 (50%) and 89 (46%) patients, respectively, for BPI-SF, and 177 (50%) and 87 (45%) patients,

	Prespecified analyses			Post-hoc analyses		
	[^{177}Lu]Lu-PSMA-617 plus standard of care (n=385)	Standard of care (n=196)	Hazard ratio (95% CI)	[^{177}Lu]Lu-PSMA-617 plus standard of care (n=385)	Standard of care (n=196)	Hazard ratio (95% CI)
FACT-P						
Total score	5·7 (4·8–6·6)	2·2 (1·8–2·8)	0·54 (0·45–0·66)	9·7 (6·9–11·5)	2·4 (1·8–4·1)	0·46 (0·35–0·61)
Pain-related subscale	4·6 (3·4–5·0)	1·8 (1·6–2·1)	0·55 (0·45–0·66)	5·9 (4·6–9·1)	1·8 (1·6–2·3)	0·55 (0·42–0·71)
Trial outcome index	6·0 (5·0–6·9)	2·2 (1·8–2·9)	0·56 (0·46–0·68)	11·0 (7·5–12·0)	2·9 (1·8–4·1)	0·48 (0·36–0·63)
Prostate cancer subscale	3·9 (3·3–4·6)	1·8 (1·6–2·3)	0·62 (0·51–0·76)	4·8 (3·4–6·4)	1·8 (1·6–2·9)	0·59 (0·46–0·76)
Physical wellbeing	4·7 (3·9–6·1)	1·8 (1·6–2·3)	0·51 (0·42–0·62)	7·6 (4·8–10·2)	1·8 (1·6–2·9)	0·44 (0·34–0·58)
Social or family wellbeing	5·9 (5·1–6·9)	2·9 (2·3–3·5)	0·55 (0·45–0·67)	14·3 (8·8–NE)	7·4 (5·8–NE)	0·83 (0·58–1·17)
Emotional wellbeing	6·6 (5·7–7·3)	2·4 (1·9–3·0)	0·52 (0·43–0·63)	11·4 (8·9–16·4)	4·4 (2·4–6·2)	0·47 (0·35–0·64)
Functional wellbeing	4·6 (3·3–5·2)	2·0 (1·6–2·6)	0·61 (0·51–0·74)	6·4 (4·4–8·5)	2·1 (1·6–3·0)	0·53 (0·41–0·69)
BPI-SF						
Pain intensity	6·9 (6·0–7·7)	2·6 (2·1–3·1)	0·52 (0·42–0·63)	14·3 (9·0–NE)	2·9 (2·0–5·5)	0·45 (0·33–0·60)
Pain interference	5·9 (5·0–6·8)	2·9 (2·3–3·5)	0·62 (0·51–0·75)	9·5 (6·5–14·4)	4·4 (1·7–6·4)	0·60 (0·45–0·80)
Worst pain intensity	5·0 (4·2–5·9)	2·0 (1·7–2·2)	0·51 (0·42–0·63)	11·3 (7·2–14·0)	2·3 (1·7–3·8)	0·49 (0·37–0·65)
EQ-5D-5L						
Utility score	1·0 (0·7–1·8)	0·5 (0·4–1·0)	0·65 (0·54–0·78)	0·9 (0·7–1·7)	0·5 (0·4–0·5)	0·49 (0·40–0·62)

Data are median (95% CI), unless stated otherwise. Time to worsening estimated from censored (symptomatic skeletal event or death) Kaplan-Meier curves. Analyses in 581 randomly assigned patients were implemented on or after March 5, 2019. BPI-SF=Brief Pain Inventory-Short Form. FACT-P=Functional Assessment of Cancer Therapy-Prostate. Lu=lutetium. NE=not estimable. PSMA=prostate-specific membrane antigen.

Table 1: **Analysis of time to worsening in months of patient-reported outcome scores**

respectively, for EQ-5D-5L (appendix p 3). Reasons for non-completion of PRO questionnaires are shown in the appendix (p 4). Mean PRO scores at baseline, changes from baseline, and the difference between treatment groups in least squares mean PRO scores from general linear models at each study visit are summarised in the appendix (pp 5–9).

Primary endpoint results (radiographic progression-free survival and overall survival) were reported previously.[12] The time to worsening in FACT-P score and all subscales was delayed in the [177Lu]Lu-PSMA-617 group compared with the control group, in both prespecified and post-hoc analyses (figure 2A, B; table 1). The median follow-up time for FACT-P score was 4·37 months (IQR 1·02–8·80) in the [177Lu]Lu-PSMA-617 group and 0·76 months (0·15–2·71) in the control group.

The time to worsening in all BPI-SF scales was delayed in the [177Lu]Lu-PSMA-617 group compared with the control group, in both prespecified and post-hoc analyses (figure 2C, D; table 1). Results of post-hoc analyses of time to worsening in BPI-SF scales using a single threshold of an increase of at least 2 points from baseline were similar to results from analyses using the prespecified threshold of an increase of at least 30% or at least 2 points from baseline (figure 2C, D; appendix p 10). The number of patients with increases of at least 30% or at least 2 points from baseline, or both, and the reasons for censoring are shown in the appendix (p 11). The median follow-up time for BPI-SF pain intensity scale was 4·14 months (IQR 1·64–8·77) in the [177Lu]Lu-PSMA-617 group and 0·66 months (0·20–2·56) in the control group. The time to improvement after worsening in BPI-SF pain intensity scale was shorter in the [177Lu]Lu-PSMA-617 group than the control group (appendix p 12).

The time to worsening in EQ-5D-5L utility score was delayed in the [177Lu]Lu-PSMA-617 group compared with the control group, in both prespecified and post-hoc analyses (figure 2E, F; table 1). In post-hoc analyses, the time to worsening in EQ-5D-5L utility score of 0·10 points was also delayed in the [177Lu]Lu-PSMA-617 group compared with the control group, when clinical disease progression or death were included or excluded from the definition (appendix p 13). EQ-VAS values by visit are summarised in the appendix (p 14).

Symptomatic skeletal events occurred in 60 (16%) of 385 patients in the [177Lu]Lu-PSMA-617 group and 34 (17%) of 196 patients in the control group (table 2). The predominant type of symptomatic skeletal event was radiation to relieve bone pain and spinal cord compression. The median follow-up time for symptomatic skeletal events was similar between the treatment groups (table 3). In post-hoc subgroup analsyes, among 310 (53%) of 581 patients who did not receive bone-targeted therapy, the median time to first symptomatic skeletal event or death was delayed in the [177Lu]Lu-PSMA-617 group compared with the control group (table 3; appendix p 1). Among 271 (47%) of 581 patients who received bone-targeted therapy as part of standard of care at baseline, the median time to first symptomatic

	[177Lu]Lu-PSMA-617 plus standard of care (n=385)	Standard of care (n=196)
Number of events	256 (66%)	137 (70%)
Symptomatic skeletal events*	60 (16%)	34 (17%)
Symptomatic pathological bone fracture	13 (3%)	1 (1%)
Spinal cord compression	5 (1%)	11 (6%)
Radiation to relieve bone pain	42 (11%)	21 (11%)
Tumour-related orthopaedic surgical intervention	1 (<1%)	1 (1%)
Deaths	196 (51%)	103 (53%)
Number of patients censored	129 (34%)	59 (30%)
Reason for censoring		
Censored at last study visit (on or before end of treatment visit)	129 (34%)	57 (29%)
No evaluable data (censored at randomisation)	0	2 (1%)

Data are n (%). A patient can be counted in more than one symptomatic skeletal event category (symptomatic pathological bone fracture, spinal cord compression, radiation to relieve bone pain, and tumour-related orthopaedic surgical intervention), if multiple events occurred on the same date, or subcategory (eg, location of the radiation to relieve bone pain). Analyses in 581 randomly assigned patients were implemented on or after March 5, 2019. Lu=lutetium. PSMA=prostate-specific membrane antigen. *Symptomatic skeletal events are captured on the adverse event, concurrent surgical or therapeutic procedures, and radiotherapy case report forms.

Table 2: **Types of symptomatic skeletal event and reasons for censoring**

	[177Lu]Lu-PSMA-617 plus standard of care (n=385)	Standard of care (n=196)	Hazard ratio (95% CI)
Bone-targeted agents			
Symptomatic skeletal event or death	115/175 (66%)	66/96 (69%)	··
Median time to event, months	12·0 (10·0–14·2)	7·2 (5·6–10·2)	0·49 (0·36–0·68)
Median follow-up time, months	17·1 (14·6–17·7)	16·6 (11·0–NE)	··
No bone-targeted agents			
Symptomatic skeletal event or death	141/210 (67%)	71/100 (71%)	··
Median time to event, months	11·5 (9·8–13·7)	5·8 (4·1–9·2)	0·50 (0·37–0·68)
Median follow-up time, months	17·0 (15·6–17·4)	19·8 (11·5–NE)	··
Overall			
Symptomatic skeletal event or death	256/385 (66%)	137/196 (70%)	··
Median time to event, months	11·5 (10·3–13·2)	6·8 (5·2–8·5)	0·50 (0·40–0·62)
Median follow-up time, months	17·0 (15·9–17·3)	16·9 (11·5–NE)	··

Data are n/N (%) or median (95% CI). Stratified by concurrent use of bone-targeted agents as part of standard of care. Hazard ratio based on Cox proportional hazards model stratified for lactate dehydrogenase (≤260 IU/L *vs* >260 IU/L); presence of liver metastases (yes *vs* no); Eastern Cooperative Oncology Group performance status score (0 or 1 *vs* 2); and inclusion of androgen receptor pathway inhibitor in best supportive care or standard of care at time of randomisation (yes *vs* no). Median time to event and median follow-up times estimated from Kaplan-Meier curves censored for event (symptomatic skeletal event or death). Analyses in the 581 patients randomly assigned after measures were implemented on or after March 5, 2019. NE=not estimable. Lu=lutetium. PSMA=prostate-specific membrane antigen.

Table 3: **Overall and post-hoc subgroup analysis of time to first symptomatic skeletal event or death**

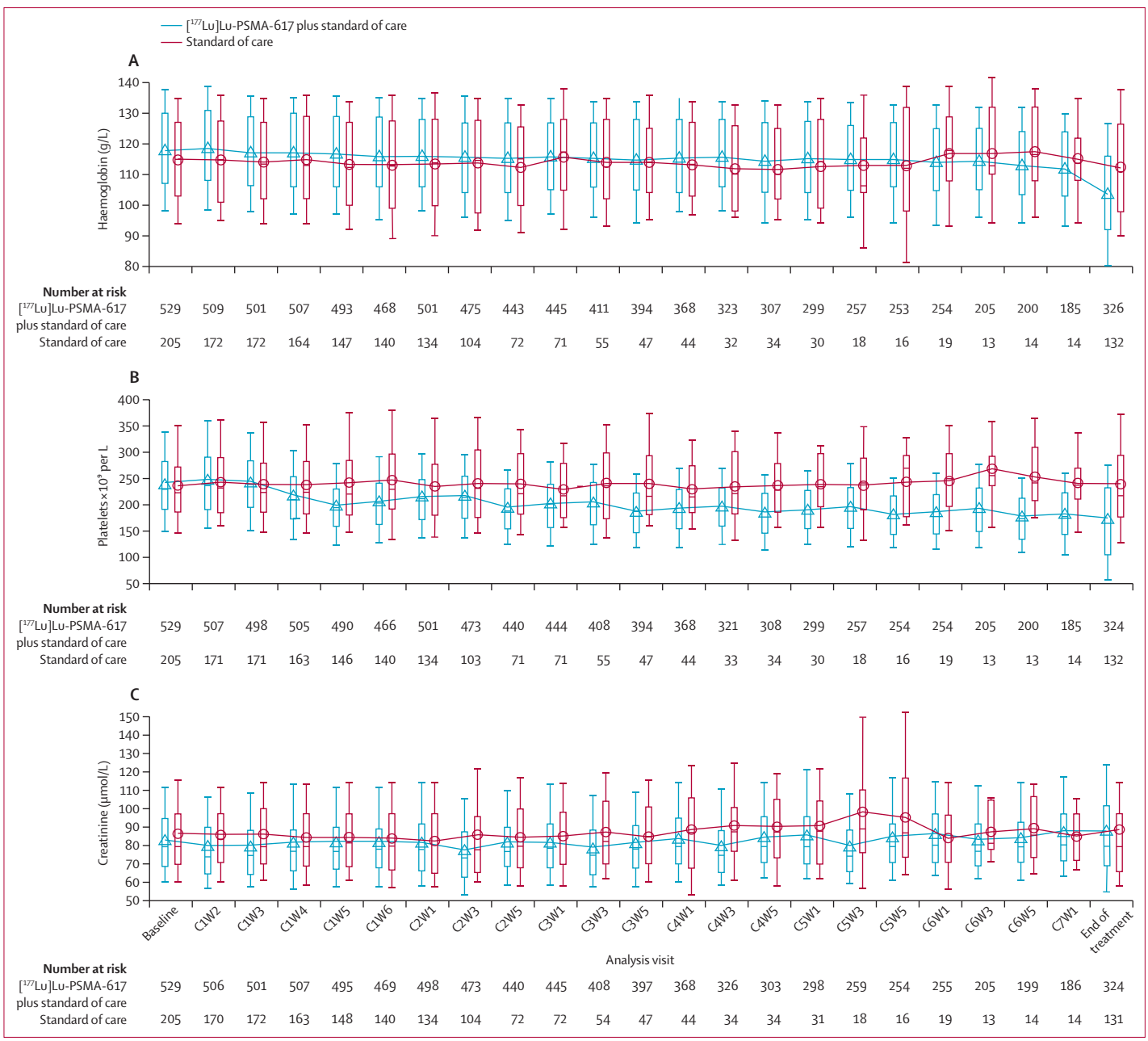


Figure 3: **Laboratory measures during treatment**

Haemoglobin (A), platelet (B), and creatinine (C) concentrations. Analyses in 734 patients who received at least one dose of study treatment. Box plot of median with 25th–75th percentiles. Whiskers extend to the 10th–90th percentiles. Triangles and circles with joint lines represent the mean values. Assessments included up to the timepoint with at least ten patients in each treatment group. Platelet values of 100 000 × 10^9/L or higher at C6W5 were not converted to the correct units and were excluded. C=cycle. Lu=lutetium. PSMA=prostate-specific membrane antigen. W=week.

skeletal event or death was also delayed in the [^{177}Lu]Lu-PSMA-617 group compared with the control group (table 3; appendix p 1). Time to first symptomatic skeletal event was also delayed in the [^{177}Lu]Lu-PSMA-617 group compared with the control group in a further post-hoc analysis of time to first symptomatic skeletal event that did not include death as an event, stratified by concurrent use of bone-targeted agents (appendix p 15).

During the treatment period, the median follow-up time in the safety analysis set was 14·78 months (IQR 9·10–19·35) in the [^{177}Lu]Lu-PSMA-617 group and 10·64 months (5·09–17·22) in the control group. The most common adverse events, occurring in 10% or more of patients in either group, were fatigue (228 [43%] of 529 patients who received [^{177}Lu]Lu-PSMA-617 plus standard of care *vs* 47 [23%] of 205 who received standard of care only), dry mouth (205 [39%] *vs* one [<1%]), nausea

(187 [35%] *vs* 34 [17%]), anaemia (168 [32%] *vs* 27 [13%]), back pain (124 [23%] *vs* 30 [15%]), arthralgia (118 [22%] *vs* 26 [13%]), decreased appetite (112 [21%] *vs* 30 [15%]), constipation (107 [20%] *vs* 23 [11%]), diarrhoea (100 [19%] *vs* six [3%]), vomiting (100 [19%] *vs* 13 [6%]), thrombocytopenia (91 [17%] *vs* nine [4%]), lymphopenia (75 [14%] *vs* eight [4%]), leukopenia (66 [12%] *vs* four [2%]), bone pain (59 [11%] *vs* 17 [8%]), urinary tract infection (58 [11%] *vs* two [1%]), decreased weight (57 [11%] *vs* 18 [9%]), and dyspnoea (53 [10%] *vs* 20 [10%]). Grade 3 or 4 haematological adverse events were more frequent in patients who received [^{177}Lu]Lu-PSMA-617 plus standard of care than those who received standard of care alone, particularly for decreased haemoglobin (80 [15%] *vs* 13 [6%]), decreased lymphocyte counts (269 [51%] *vs* 39 [19%]), and decreased platelet counts (49 [9%] *vs* five [2%]; appendix p 16). Mean haemoglobin concentrations and platelet counts remained stable over time (figure 3A–B). Haemoglobin concentrations by visit during treatment are shown in the appendix (p 17). Haemoglobin concentrations based on CTCAE grade during treatment are shown in the appendix (p 18). 21 (4%) of 529 patients had [^{177}Lu]Lu-PSMA-617 dose reductions owing to myelosuppression (appendix p 19). Treatment-related adverse events leading to death occurred in five (1%) patients in the [^{177}Lu]Lu-PSMA-617 group (pancytopenia [n=2], bone marrow failure [n=1], subdural haematoma [n=1], and intracranial haemorrhage [n=1]) versus no patients in the control group.

The incidence of increased creatinine concentrations of CTCAE grade 3 or higher was low, and was similar between patients who received [^{177}Lu]Lu-PSMA-617 plus standard of care and those who received standard of care only (appendix p 16). Mean creatinine concentrations remained stable over time (figure 3C). Data on adverse events have been previously published.[12]

Discussion

The addition of [^{177}Lu]Lu-PSMA-617 to standard of care delayed the time to worsening of HRQOL and pain in patients with metastatic castration-resistant prostate cancer who previously received at least one androgen receptor pathway inhibitor and one or two taxanes. Worsening was delayed across all PROs, subscales, and domains, with the greatest difference in median time to event in BPI-SF pain intensity and FACT-P score. Similar delays in time to worsening in score, clinical disease progression, or death were detected in prespecified and post-hoc analyses. These results complement the previously reported longer radiographic progression-free survival and extended overall survival with [^{177}Lu]Lu-PSMA-617 in the VISION trial.[12]

VISION aimed to investigate [^{177}Lu]Lu-PSMA-617 as an add-on to a broad range of standard of care therapies that could safely be combined with investigational radioligand therapy.[12] Therefore, protocol-permitted standard of care did not include chemotherapy or radium-223, and patients who were candidates for a second taxane regimen at baseline were ineligible.[12] Drugs that were investigational at the time of study design were also not included in protocol-permitted standard of care (eg, olaparib and other poly [ADP-ribose] polymerase inhibitors).[12]

Time to HRQOL worsening was delayed with [^{177}Lu]Lu-PSMA-617 on both the FACT-P and EQ-5D-5L. For the FACT-P total score, the threshold for worsening with a decrease of at least 10 points from baseline is well documented.[16] Unlike the prostate cancer-specific FACT-P instrument, the EQ-5D-5L is generic, with no established cutoff and potentially low sensitivity in patients with advanced prostate cancer.[17] Although previous studies have used the EQ-5D-5L to complement prostate-cancer specific instruments, patients were at an earlier stage of disease than in the extensively pretreated VISION population.[19,20] Owing to the potentially low sensitivity of EQ-5D-5L in the study population, worsening was predefined as no change or any deterioration in utility score from baseline, leading to rapid worsening. However, deterioration was also rapid using a post-hoc EQ-5D-5L worsening threshold of 0·10 points in utility score. These results are different to those obtained using the more reliable and robust FACT-P and BPI-SF, for which definitions of worsening were based on well established thresholds.[16,17]

FACT-P and BPI-SF outcomes improved with cabazitaxel treatment in patients with advanced metastatic castration-resistant prostate cancer in a prospective real world study.[21] In the AFFIRM trial[16] of enzalutamide after docetaxel, FACT-P score and subscales improved compared with placebo in patients with progressive metastatic castration-resistant prostate cancer. These findings suggest that therapies that delay disease progression are associated with maintained HRQOL. In VISION, worsening in BPI-SF scales and subscales was defined as an increase in score of at least 30% or at least 2 points from baseline, which is the established clinically meaningful threshold used in several clinical trials of patients with prostate cancer.[17] The COU-AA-302 study[22] used a slightly different definition of an increase of at least 30% from baseline at two consecutive evaluations. In VISION, most patients meeting the definition of BPI-SF worsening had at least a 2-point increase in score from baseline and post-hoc analyses of time to BPI-SF worsening confirmed results using the 2-point threshold.

Pain in patients with metastatic castration-resistant prostate cancer can be attributed largely to bone metastases, lumbosacral invasion, and nerve root compression, and is associated with shortened overall survival.[23] Bone metastases are frequent,[24] leading to pain, symptomatic skeletal events,[25] disability, impaired HRQOL,[7,26] and mortality.[27,28] Symptomatic skeletal events had a substantial negative effect on HRQOL and pain in a large cohort study in patients with castration-resistant prostate cancer and bone metastasis.[5,6] In the CARD study,[19] cabazitaxel delayed the time to first symptomatic

skeletal event and improved pain response versus abiraterone or enzalutamide in patients with metastatic castration-resistant prostate cancer, but did not delay the time to worsening in FACT-P score. In COMET-1,[4] cabozantinib delayed the time to first symptomatic skeletal event in patients with metastatic castration-resistant prostate cancer versus prednisone. In VISION,[12] the median time to first symptomatic skeletal event or death was delayed in the [^{177}Lu]Lu-PSMA-617 group versus control, and spinal cord compression—the most deleterious symptomatic skeletal event—was numerically less frequent. In previous studies, bone-targeted agents (denosumab or zoledronic acid) led to a decreased risk of fracture in patients with metastatic castration-resistant prostate cancer and bone metastasis.[29–31] In the present analyses, time to symptomatic skeletal event or death was delayed in the [^{177}Lu]Lu-PSMA-617 group compared with control, regardless of bone-targeted therapy use.

Adverse events in the myelosuppression and renal effects groupings were more frequent during treatment with [^{177}Lu]Lu-PSMA-617 plus standard of care than with standard of care alone, but were mainly low grade.[12] Haematology and clinical chemistry results also show a manageable haematological and renal safety profile. Although grade 3 or higher haematological adverse events were more frequent with [^{177}Lu]Lu-PSMA-617, mean haemoglobin concentrations and platelet counts remained stable during treatment. The incidence of grade 3 or higher creatinine adverse events was low and similar between groups, and mean concentrations remained stable. This safety profile supports [^{177}Lu]Lu-PSMA-617 use in often older, frail patients with late-stage disease[32] and possibly in younger patients with early-stage disease.

Ongoing phase 3 trials are investigating whether [^{177}Lu]Lu-PSMA-617 provides therapeutic benefit earlier in the treatment sequence than in our study. PSMAfore (NCT04689828) is assessing the efficacy and safety of [^{177}Lu]Lu-PSMA-617 versus change of androgen receptor pathway inhibitor in taxane-naive patients with metastatic castration-resistant prostate cancer. PSMAddition (NCT04720157) is assessing the efficacy and safety of [^{177}Lu]Lu-PSMA-617 plus standard of care versus standard of care alone in patients with metastatic hormone-sensitive prostate cancer. SPLASH (NCT04647526) is assessing the efficacy and safety of another PSMA-targeted radioligand therapy known as ^{177}Lu-PSMA-I&T versus abiraterone or enzalutamide in patients with metastatic castration-resistant prostate cancer. ECLIPSE (NCT05204927) will compare [^{177}Lu]Lu-PSMA-I&T with hormone therapy in patients with metastatic castration-resistant prostate cancer. All these studies, except SPLASH, will also investigate HRQOL or pain, or both, as PROs.

PRO questionnaires were completed during study treatment, but not after disease progression or treatment discontinuation, which is an important limitation of our study because potential subsequent effects of study treatment on HRQOL and pain were not captured. This limitation is shared by many phase 3 trials in patients with castration-resistant prostate cancer. The VISION population was heavily pretreated and had short life expectancy, so deterioration in HRQOL and pain was likely to occur within the follow-up time. Death and disease progression were included in the prespecified composite endpoints of HRQOL and pain worsening, which were assumed to have worsened in patients who did not complete questionnaires because of death or disease progression. The results support the conclusion that there was no adverse effect of [^{177}Lu]Lu-PSMA-617 on HRQOL or pain, despite the limitation that data were collected only during study treatment. Another limitation was the exclusion of chemotherapy and systemic radiotherapy from standard of care because the safety profile of these has not been established in combination with [^{177}Lu]Lu-PSMA-617.

PRO completion rates were similar in the [^{177}Lu]Lu-PSMA-617 and control groups among patients remaining on study treatment. The proportion of patients remaining in the study was smaller in the control group than the [^{177}Lu]Lu-PSMA-617 group owing to the higher dropout rate, especially after cycle three (because the treatment period was shorter in the control group than the [^{177}Lu]Lu-PSMA-617 group) and PRO questionnaires were not completed after discontinuation. This limitation means the results should be interpreted with caution. Furthermore, the cycle length was 6 weeks for cycles one to six of standard of care with or without [^{177}Lu]Lu-PSMA-617 and 12 weeks for subsequent cycles of standard of care. Sustained worsening was not assessed because of the long interval between PRO assessments (6 or 12 weeks), and because of inevitable declining HRQOL and death in patients with advanced and highly pretreated metastatic castration-resistant prostate cancer. Random assignment to open-label standard of care alone might have adversely affected patients' HRQOL. The absence of double-blinding and placebo control are limitations shared by studies of other radiopharmaceuticals.[33] The effect of a high early dropout rate in the control group was mitigated by analysing HRQOL and pain outcomes in the PFS-FAS, which included patients who were randomly assigned after implementation of measures that reduced the dropout rate in the control group.[12] Collecting HRQOL and pain data for as many patients as possible is important to replicate our findings in future trials.

Treatment with [^{177}Lu]Lu-PSMA-617 plus standard of care delayed radiographic progression-free survival and extended overall survival in patients with metastatic castration-resistant prostate cancer[12] and was associated with a longer period without deterioration in patient-reported HRQOL and pain than seen with standard of care alone. Low rates of haematological and renal toxicity were observed with [^{177}Lu]Lu-PSMA-617. These findings strengthen the rationale for use of [^{177}Lu]Lu-PSMA-617 as

a treatment option in patients with metastatic castration-resistant prostate cancer who received previous androgen receptor pathway inhibitor and taxane treatment.

Contributors

KF, KH, BJK, KR, KNC, MJM, OS, STT, RAM, and JdB conceptualised and designed the study. MD was responsible for data curation and formal analysis. KF, KH, BJK, KR, KNC, MJM, OS, STT, ATK, NV, JC, JN, XXW, VSK, J-MB, BC, and JdB were responsible for conducting the study locally. KF and MDS have accessed and verified all the data in the study. All authors contributed to drafting and critically reviewing the manuscript, had full access to all the data in the study, and accept responsibility to submit for publication.

Declaration of interests

KF reports consultant or advisory fees from Janssen, Bayer, Astellas, Sanofi, Orion Pharma, CureVac, AstraZeneca, ESSA Pharma, Amgen, Bristol Myers Squibb, Clovis Oncology, Lilly, Novartis, Daiichi-Sankyo, Pfizer, and MSD. KH reports consultant or advisory fees from ABX, AstraZeneca, Advanced Accelerator Applications (Novartis), Aktis Oncology, Amgen, Bayer, BTG, Curium, Debiopharm, Endocyte, GE HealthCare, Janssen, Ipsen, Novartis, Pharma15, Siemens Healthineers, Sirtex Medical, SOFIE Biosciences, Theragnostics, and Y-mAbs Therapeutics. BJK reports consultant or advisory fees from Advanced Accelerator Applications (Novartis), Astellas, Bayer, Janssen, Terumo, PSFI, and Isotope Technologies Munich; and research funding from Advanced Accelerator Applications (Novartis). KR reports consulting or advisory fees from ABX-CRO, ABX, Advanced Accelerator Applications (Novartis), and Bayer. KNC reports consultant or advisory fees from AstraZeneca, ESSA Pharma, Janssen, Merck, POINT Biopharma, and Roche; and research funding from AstraZeneca, ESSA Pharma, Janssen, Merck, POINT Biopharma, Novartis, Roche, and Arvinas. MJM reports consultant or advisory fees from Exelixis, Lanctheus, AstraZeneca, Amgen, Daiichi, Convergent Therapeutics, Clarity Pharmaceuticals, Pfizer, Blue Earth Diagnostics, and Isotope Technologies Munich; and research funding from Novartis. OS reports consultant or advisory fees from Advanced Accelerator Applications (Novartis), Astellas, AstraZeneca, Bayer, Blue Earth Diagnostics, Bavarian Nordic, Bristol Myers Squibb, Clarity Pharmaceuticals, Clovis Oncology, Constellation, Dendreon, EMD Serono, Fusion, Isotope Technologies Munich, Merck, Janssen, Myovant, Myriad, Noria Therapeutics, NorthStar, Novartis, Noxopharm, Progenics, POINT Biopharma, Pfizer, Sanofi, Tenebio, Telix, and Theragnostics; research funding from Bayer, Endocyte, Merck, InVitae, Constellation, Amgen, Advanced Accelerator Applications (Novartis), Arvinas, AstraZeneca, Dendreon, Janssen, Lantheus, Tenebio, and Progenics; holds an issued US patent (number 7 166 691); and owns stock in Lilly, GlaxoSmithKline, AbbVie, Cardinal Health, UnitedHealth Group, Clarity Pharmaceuticals, Noria Therapeutics, and Clovis Oncology. STT reports consultant or advisory fees from Sanofi, Medivation, Astellas, Janssen, Genentech, Bayer, Eisai, Abbvie, Tolmar, Seattle Genetics, Amgen, Clovis Oncology, Pfizer, Novartis, Clarity Pharmaceuticals, Genomic Health, POINT Biopharma, Blue Earth Diagnostics, Aikido, Telix, Convergent Therapeutics, EMD Serono, Myovant, and Merck; research funding from Sanofi, Medivation, Astellas, Janssen, Amgen, Genentech, Newlink, Bristol-Myers Squibb, Inovio, AstraZeneca, Aveo, Rexahn, Bayer, Merck, Abbvie, Karyopharm, Endocyte, Clovis Oncology, Seattle Genetics, Novartis, Gilead, and POINT Biopharma; holds a patent pending with Gilead; and owns stock in Alkido. NV reports consultant or advisory fees from Genzyme, MSD, Janssen, Eisai, AstraZeneca, Dendreon, Pfizer, Bayer, Seagen, Clovis Oncology, Advanced Accelerator Applications (Novartis), and Amgen. JC reports consultant or advisory fees from Astellas, Blue Earth Diagnostics, Curium, DS Pharma, GE Healthcare, Isoray, Janssen, Lightpoint Medical, Lantheus, POINT Biopharma, Radiomedix, IBA RadioPharma, Monrol, Novartis, Telix, and Sanofi; and research funding from Lantheus, Novartis, and POINT Biopharma. JN reports consultant or advisory fees from Curium, POINT Biopharma, Bayer, and Pfizer; and research funding from Advanced Accelerator Applications (Novartis) and ABX. ATK and XXW report consultant or advisory fees from Advanced Accelerator Applications (Novartis). VSK reports consultant or advisory fees from Clovis Oncology, Pfizer, EMD Serono, Seagen, AstraZeneca, Janssen, Astellas, Seattle Genetics, Dendreon, Guidepoint, GLG, and ExpertConnect; and research funding from Janssen, Clovis Oncology, Nektar, Taiho, Merck, and Advanced Accelerator Applications (Novartis). JdB reports consultant or advisory fees from AstraZeneca, Astellas, Bayer, Cell Centric, Daiichi, Genentech (Roche), Endocyte, Advanced Accelerator Applications (Novartis), Pfizer, GSK, Janssen, Merck Serono, MSD, Sanofi Aventis, and Pfizer; and research funding from AstraZeneca, Bayer, CellCentric, Daiichi, GSK, Harpoon, Janssen, Merck Serono, MSD, Pfizer, Sanofi Aventis, and Genentech (Roche). RG, MD, and RAM are employees or own Novartis stocks and shares. J-MB and BC declare no competing interests.

Data sharing

Advanced Accelerator Applications (Novartis) is committed to sharing patient-level data and supporting clinical documents with bona fide researchers at request to the corresponding author. The process and data availability for this study are available when criteria are met.

For the **data request criteria** see https://www.clinicalstudydatarequest.com/

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For the **Good Publication Practice guidelines** see https://www.ismpp.org/gpp3

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